

# Airframe Technology Master

## FATIGUE & DT

### Lecture 03

## Fatigue Endurance. Crack Initiation

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## 10.3 Basic Concept

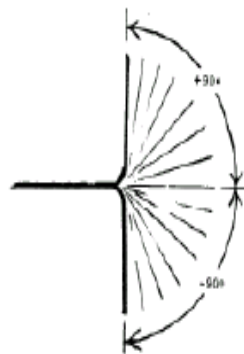
## Basic Fatigue concept

The knowledge of the materials behaviour is essential to understand the Fatigue phenomena. First important thing to know is the microstructure of materials, the basics of crystalline structures, such as the theory of dislocations, sliding planes and other concepts.

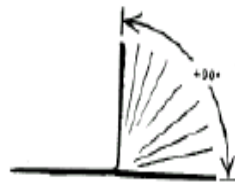
If a load is gradually applied to a piece, it finally breaks with a load that is known as the “static ultimate load”. If the same piece is subject to a fraction of that ultimate load, the piece does not break, this load just produces strains. If there are no permanent strains when the load disappears, the piece recovers its original shape. However, something occurred, because if this load cycle is applied again a given number of times, it finally breaks, after the initiation and further growth of a crack. This breakage phenomenon after a repeating application of a series of load cycles with a lower magnitude than ultimate one is known as Fatigue.

## Basic Fatigue concept

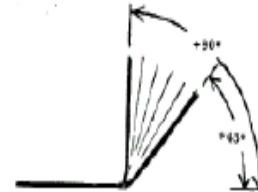
Some simple examples can easily demonstrate the basic principles of metal Fatigue. For instance, everybody has experienced the difficulty of breaking a galvanized steel wire by a single manual pull, but it can be easily broken by folding it four or five times (cycles) of  $-90^\circ$  to  $+90^\circ$ . If it is folded from  $0^\circ$  to  $90^\circ$ , it will break in 15 or 16 cycles, if it is folded from  $45^\circ$  to  $90^\circ$  it will break in 60 times.



5 Cycles



15 Cycles



60 Cycles

## Basic Fatigue concept

As we verified in the previous example, the magnitude of the applied load is not the only influencing factor, but the way to apply it.

A clarifying example is this: if a rod with a ultimate load of 524 KN is subjected to load cycles of  $-240$  to  $+240$  KN, it will break at 39.000 cycles. If the cycles are from 0 to  $+240$  KN, the rod would break at 355.000 cycles approximately, and if they are from 0 to  $+345$  KN, the rod would break at 56.000 cycles.

If the breakage surface of the piece is carefully observed, the typical fatigue marks that indicate the place where the cracks are formed and the further growth due to the application of the load cycles would be observed. Therefore, it is important to analyse the breakage surfaces to determine if Fatigue problems occurred before the breakage.

As it will be observed later, there are other parameters that influence the occurrence of Fatigue problems, such as the piece geometry, material used, etc.

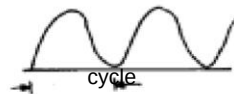
## Basic Fatigue concept



524 KN  
STATIC  
BREAKAGE



345 KN  
(TENSILE)



BREAKAGE AT  
56.000 CYCLES



240 KN (TENSILE)

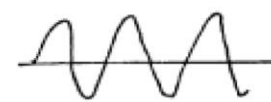


BREAKAGE AT  
355.000 CYCLES



CRACK INITIATION  
FATIGUE BEACHES

240 KN (TENSILE)

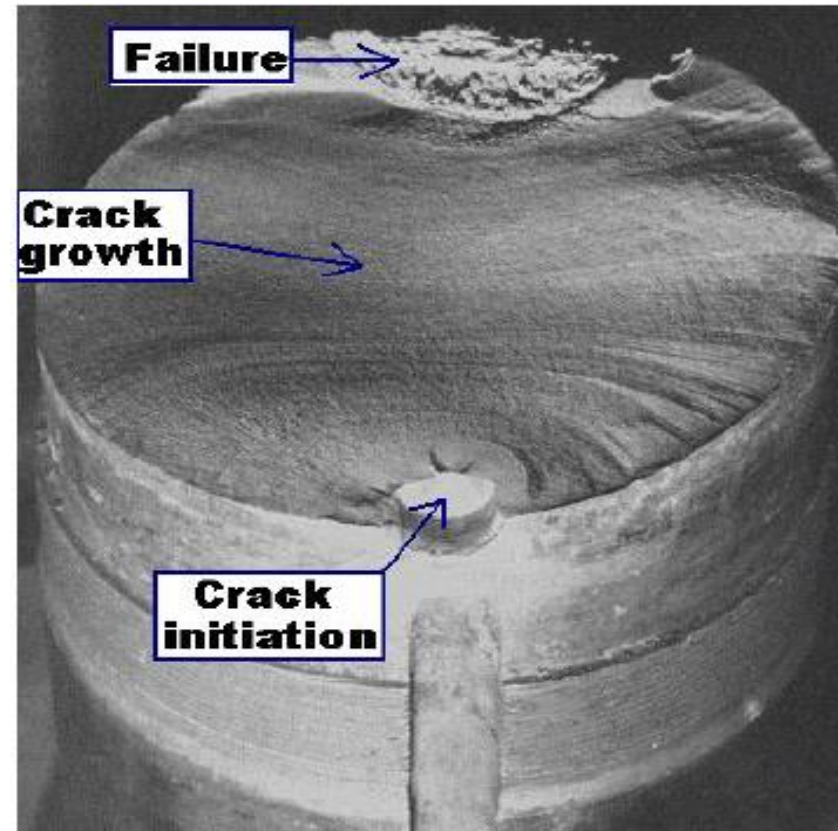


240 KN  
(COMPRESSION)  
BREAKAGE AT  
39.000 CYCLES

## Basic Fatigue concept

Typical fatigue failure of a metallic crankshaft

- Crack began at the stress concentration at the hole (front of picture)
- Steps in the fracture were caused by successive cycles of loading
- The final failure took place suddenly, as shown by the small fracture surface at the rear

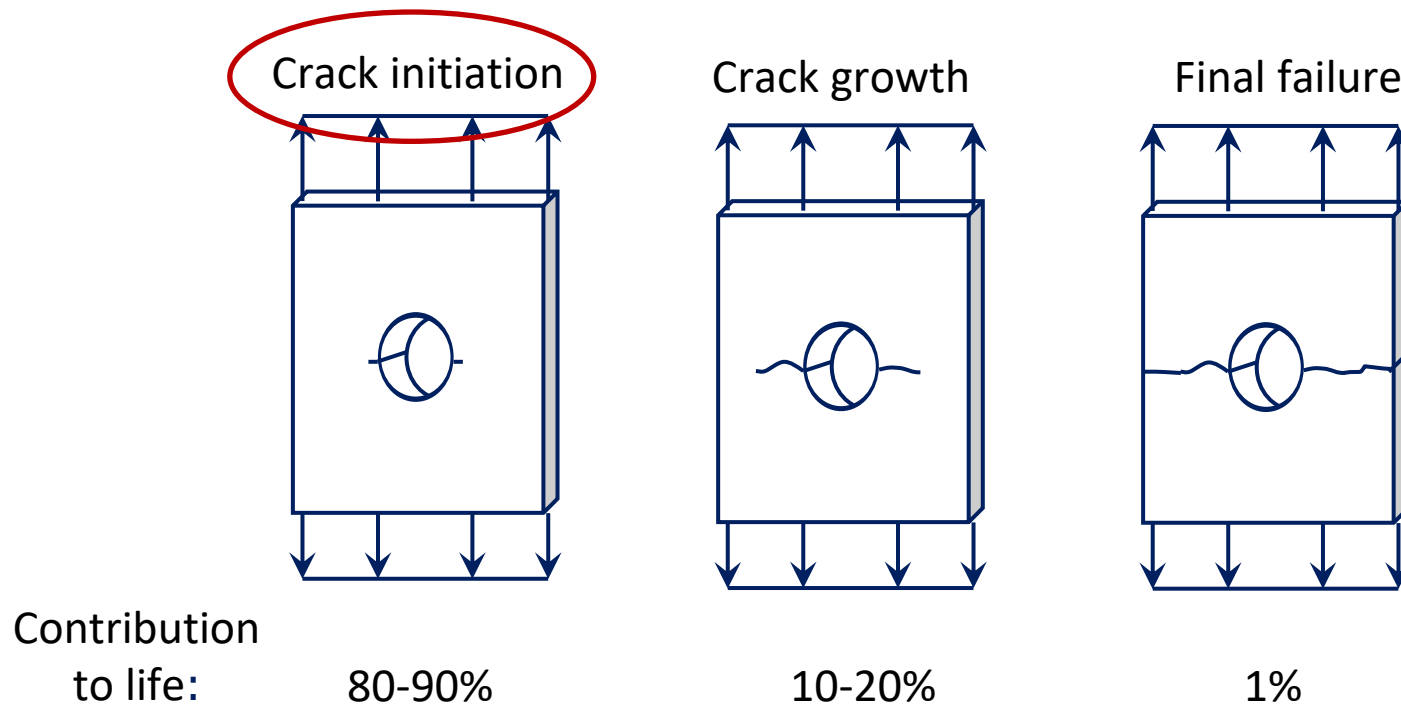




## 10.3 Crack Initiation

## Introduction

Fatigue consists in 3 phases



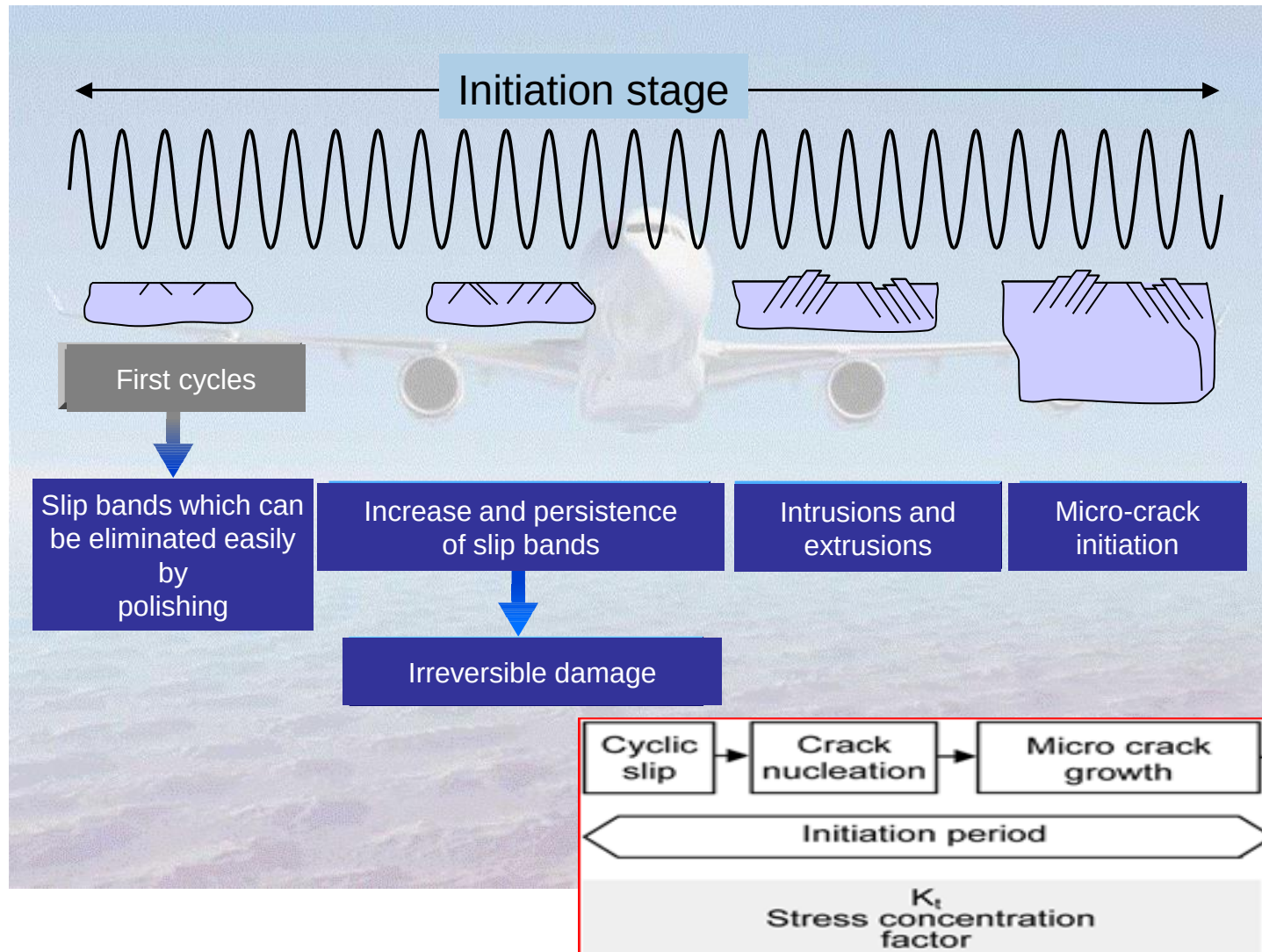
Fatigue crack initiation  $\neq$  Fatigue crack growth

## Introduction

The crack initiation process in a material subjected to a spectrum of stress cycles is produced in the following stages:

- First cycles. The first sliding slots appear, at a very superficial level; they can be easily eliminated by polishing the surface
- The sliding slots grow and become more persistent. An irreversible damage occurs
- Intrusions and extrusions. The material surface becomes more irregular while the sliding slots move
- A micro crack appears between two sliding slots due to internal tensions

## Introduction



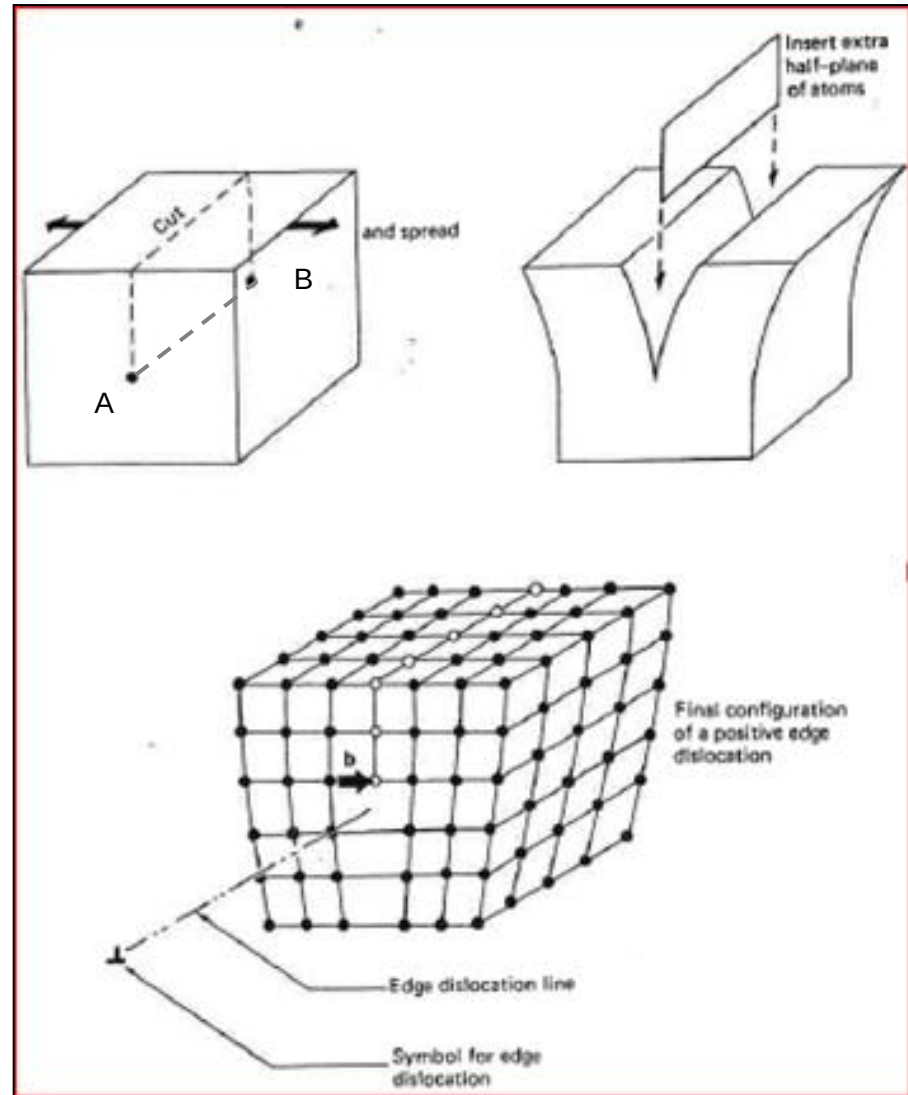
## Types of dislocations

An edge dislocation can appear as if an additional half-plane was inserted.

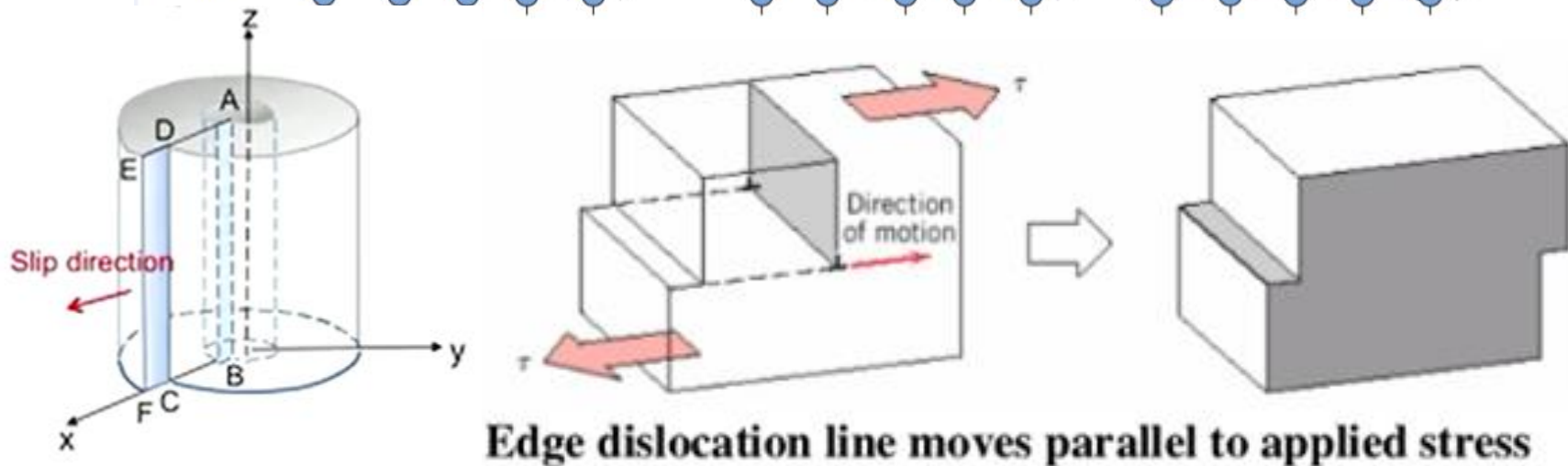
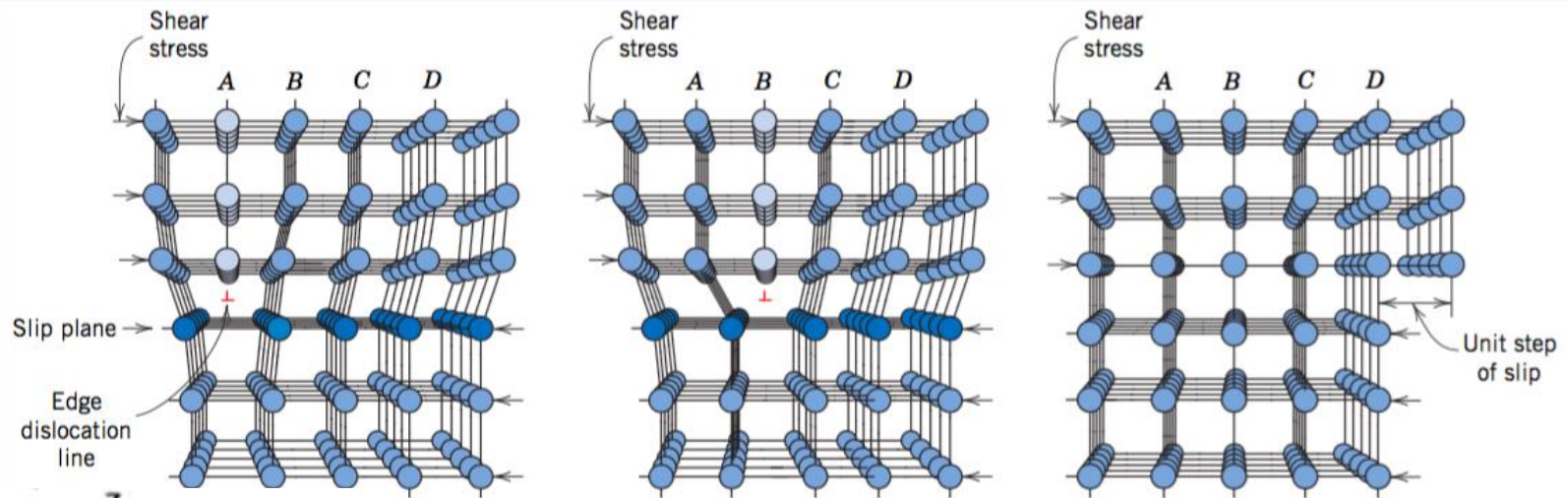
The dislocation line AB represents the axis of the cylinder inside which the crystal arrangement is disturbed.

The network is practically perfect outside this dislocation line.

<https://www.youtube.com/watch?v=ufwWl8jGG8M>



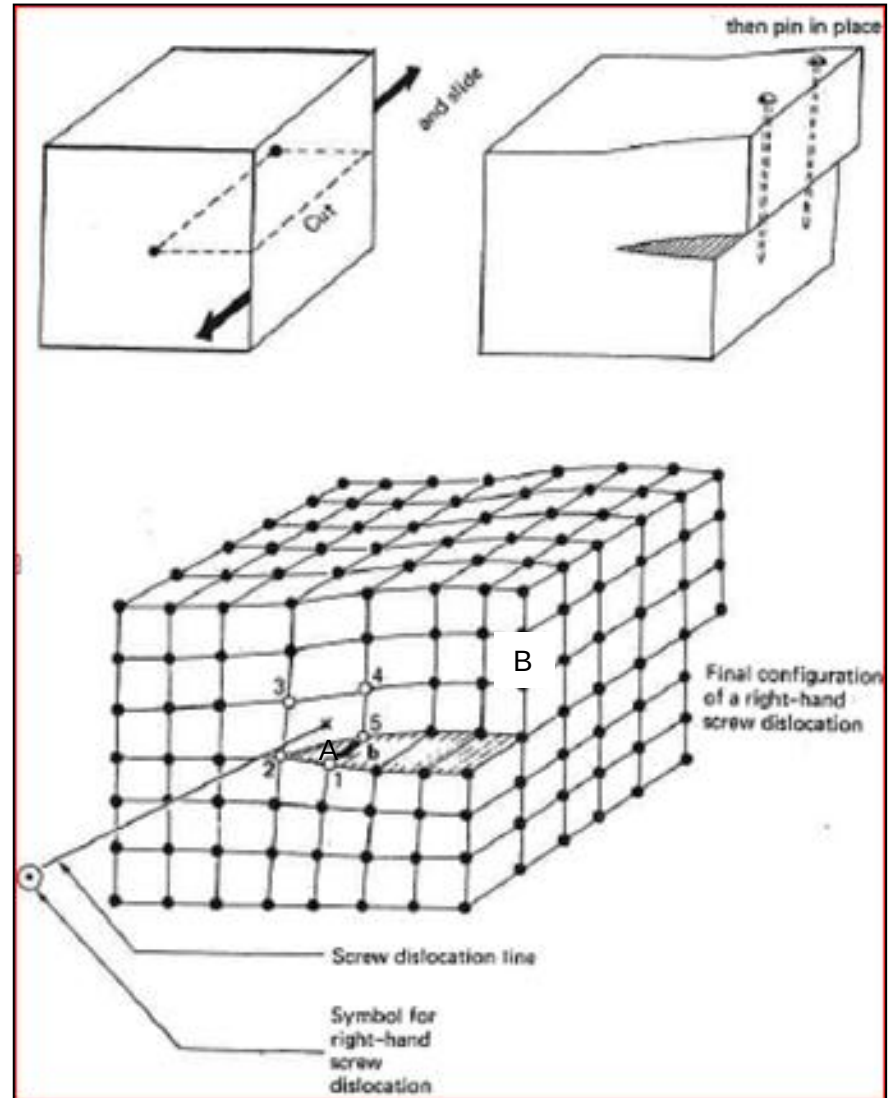
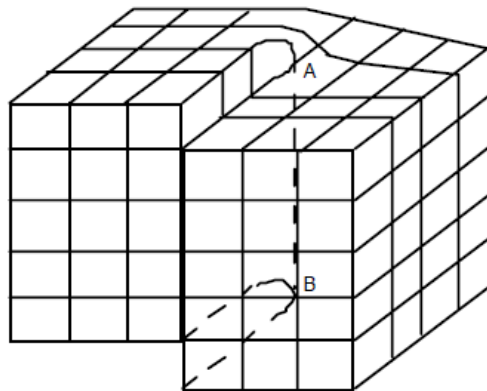
## Types of dislocations



## Types of dislocations

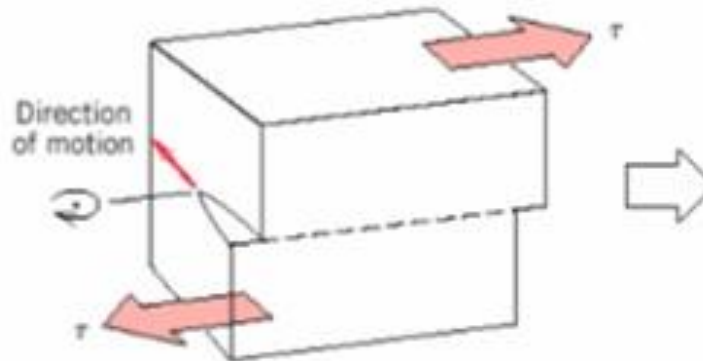
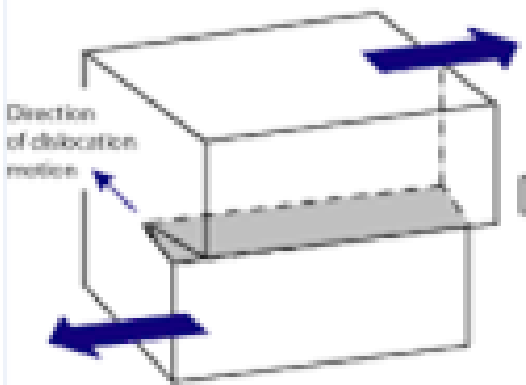
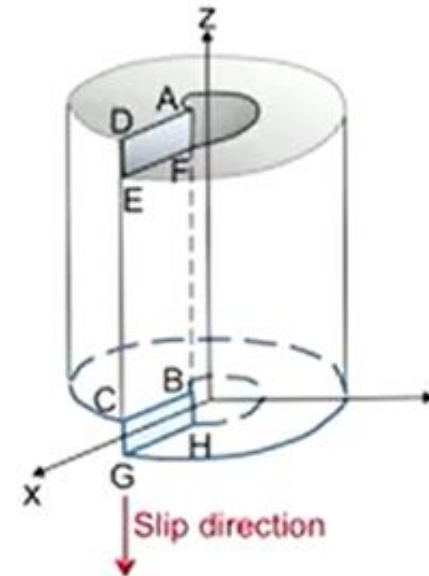
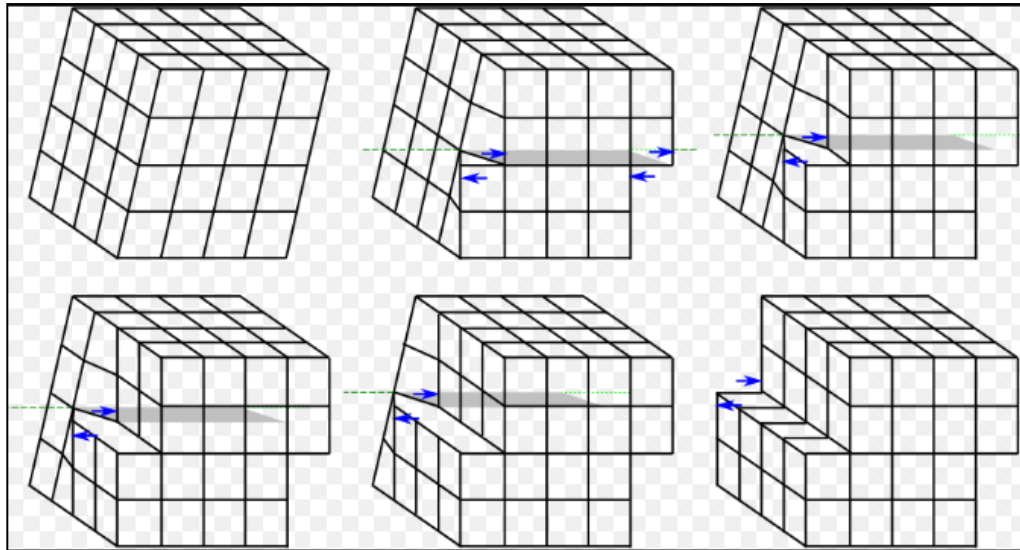
A screw dislocation is characterized as follows:

A part of the crystal network is sheared in relation to the other part: the atomic planes perpendicular to the dislocation line AB are thus transformed into a helicoidal plane

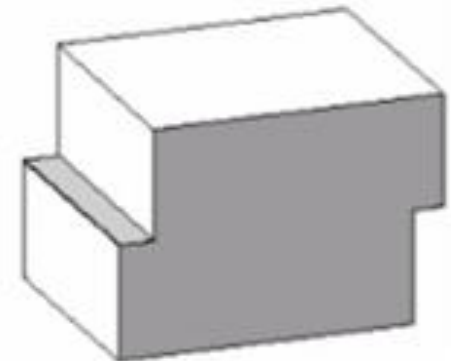




## Types of dislocations



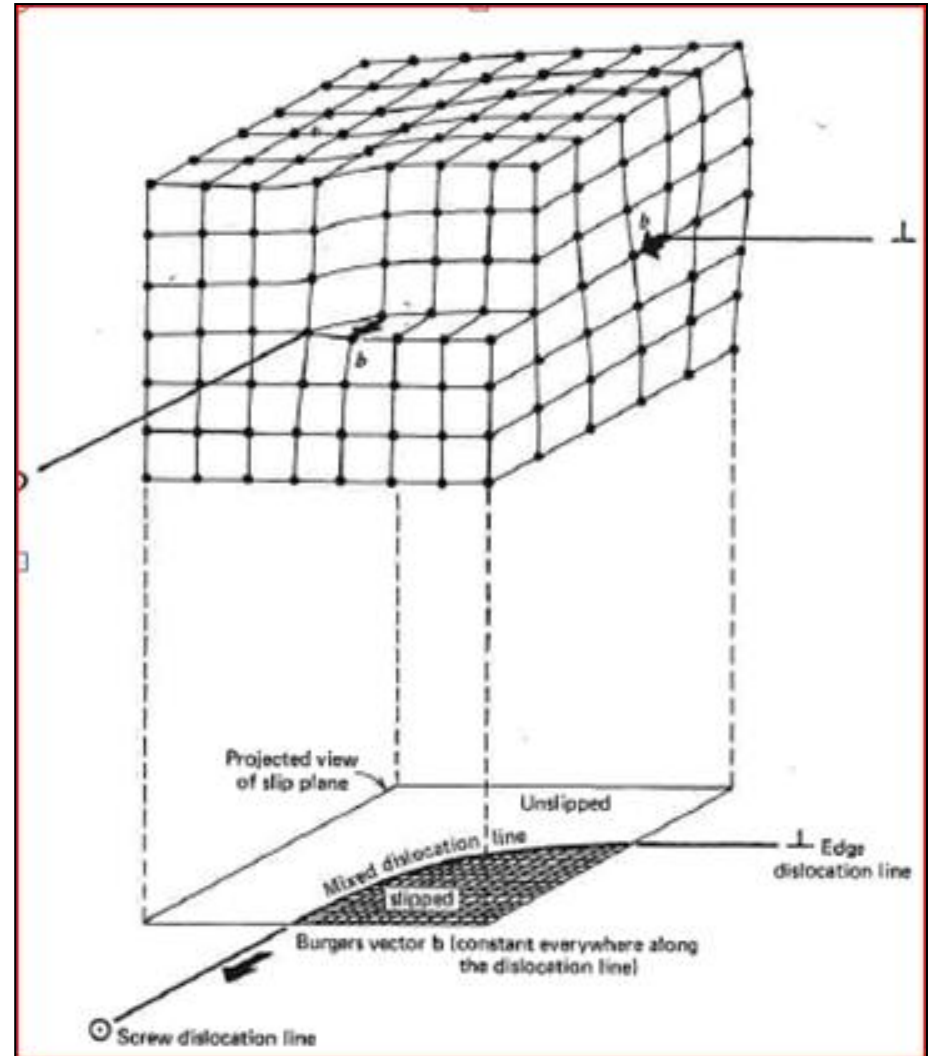
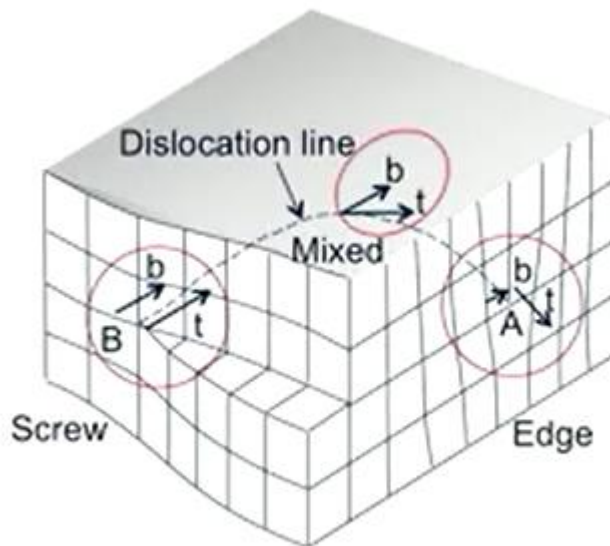
**Screw dislocation line moves perpendicular to applied stress**





## Types of dislocations

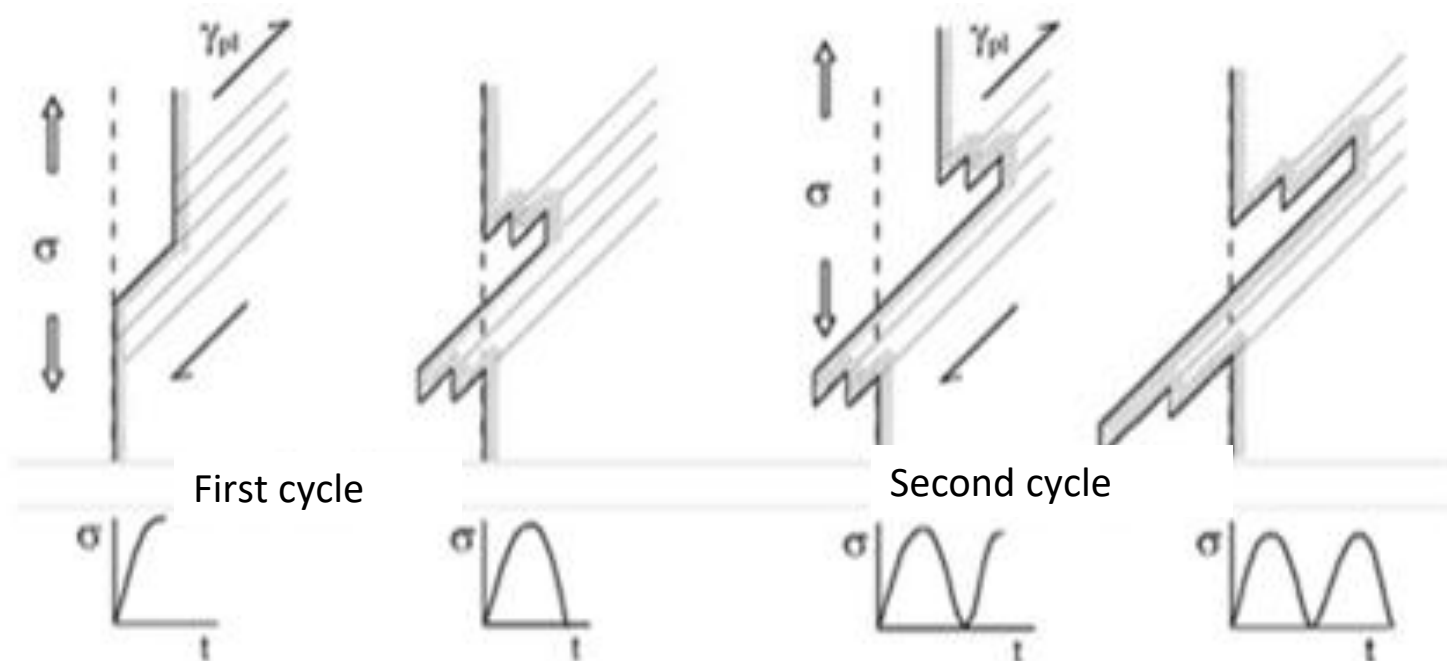
Mixed dislocation: Edge +  
Screw dislocations



## Micro-crack apparition

The displacement of the dislocations leads to permanent deformations: plasticity at macroscopic level. Certain grains of the microstructure poorly oriented in relation to the local loading can plastify.

This forms **intrusions and extrusions** on the surface of the metal which lead to **micro-cracks**.

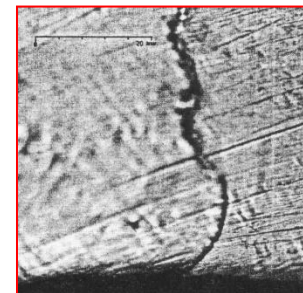
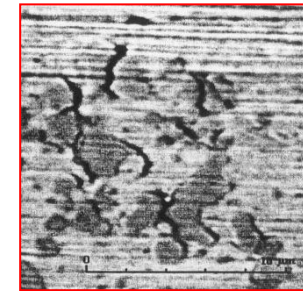
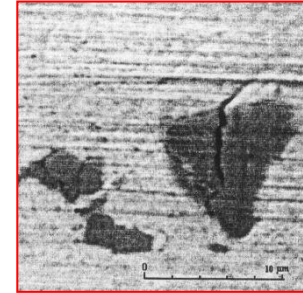


## Micro-crack apparition

When the number of cycles applied is sufficiently high, a certain number of grains are completely traversed by such micro-cracks. This phase is called the micro-crack nucleation phase.

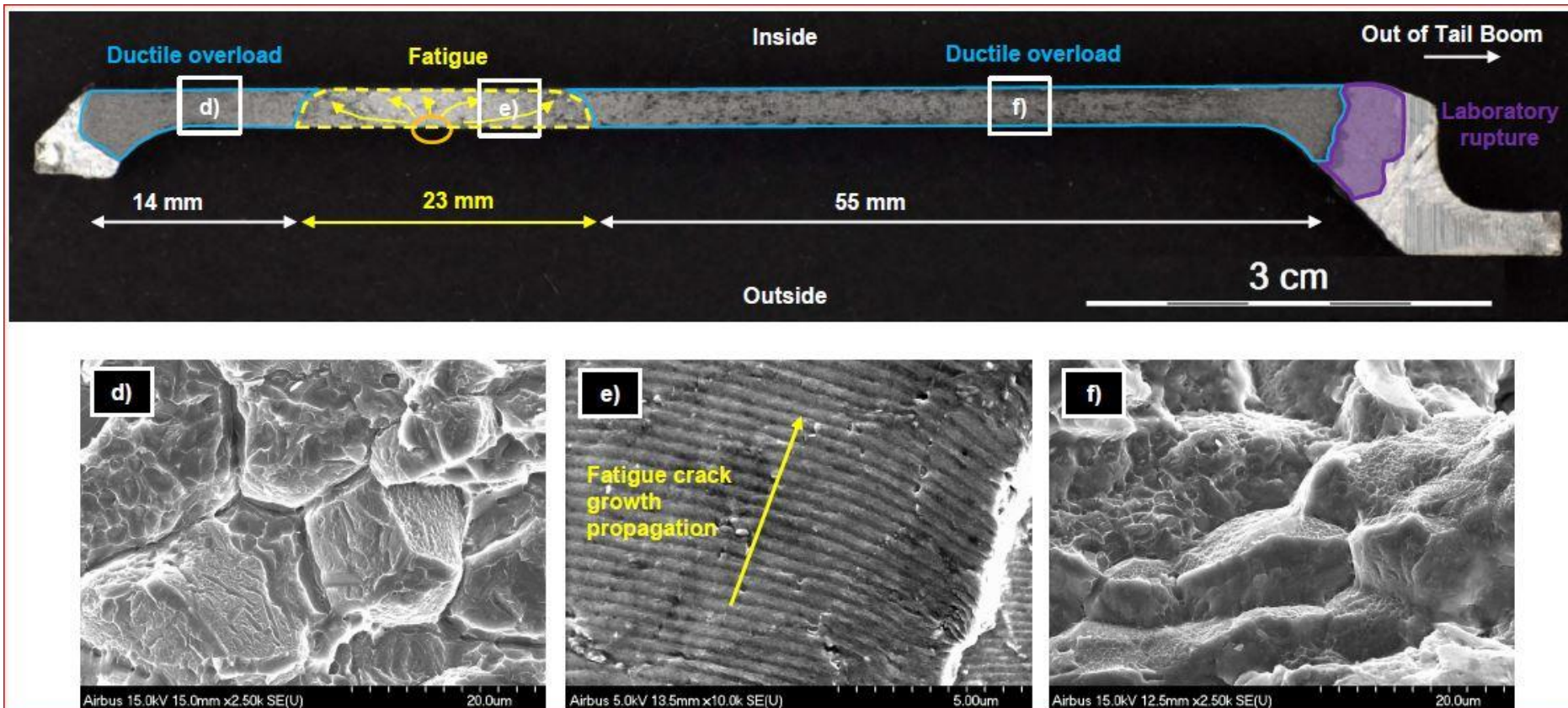
Then, these micro-cracks may join up to form a macroscopic crack oriented more or less in direction of maximum shear (micro-crack coalescence phase).

Beyond a critical size, the crack propagates in a plane perpendicular to the direction of the main maximum tensile stress and therefore changes direction (crack propagation phase). The start of this phase is generally defined as “crack initiation” in an engineering sense.



## Examples

This pictures shows the initiation of a crack at a point were there was a discontinuity producing a local stress concentration and the subsequent growth. The picture e) shows the typical striation marks characteristic of the fatigue growth.



## 10.3 Form of Fatigue

## Forms of fatigue

- Cyclic fatigue due to repeated and fluctuating loads
- Corrosion fatigue accelerated by a corrosive environment
- Fretting fatigue which is initiated by the small-scale rubbing movements and abrasion of adjacent parts
- Thermal fatigue produced by stress fluctuations induced by thermal expansions and contractions
- Acoustic fatigue caused by high frequency stress fluctuations induced by vibration excited by jet or propeller noise, etc.





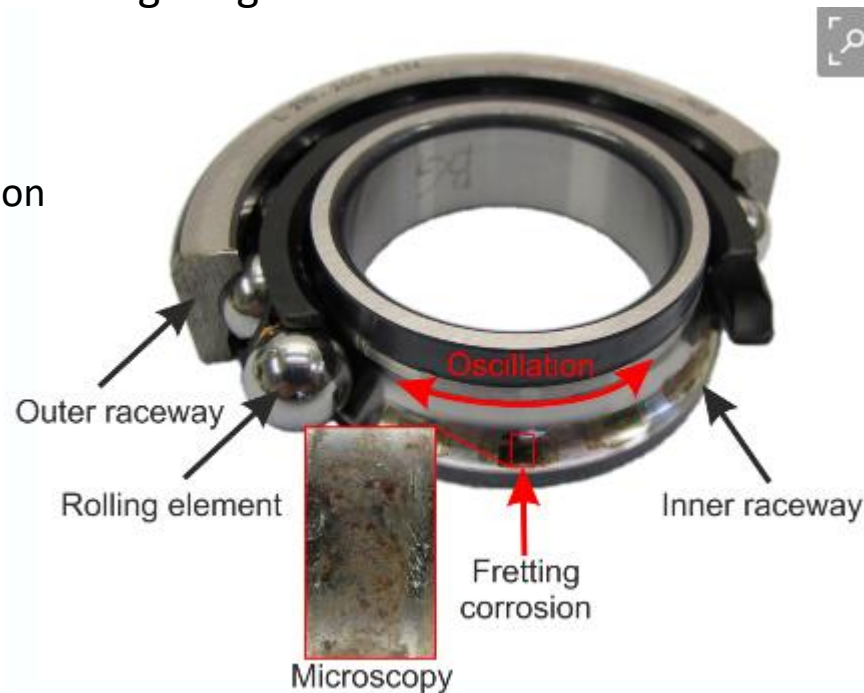
## Fatigue corrosion

- Corrosion fatigue is fatigue in a corrosive environment.
- It is the mechanical degradation of material under the joint of corrosion and cyclic loading.
- When metal is subjected to repeated mechanical stress and chemical attack, it becomes fatigued and separates, severe pitting, cracking and separation along stress lines are evidence of fatigue corrosion.
- Control of corrosion can be accomplished by either lowering the cycle stress or by corrosion control.



## Fretting fatigue

- A special wear process that occurs at the contact area between two materials under load and subject to small relative motion by vibration or some other forces.
- Factors which influence fretting fatigue most are:
  - Temperature
  - Material in contact
  - Cyclic frequencies
  - Environmental condition
  - Contact surfaces
  - Coefficient of friction
  - Hardness of material
  - Contact pressure





## Acoustic/Sonic fatigue

**The acoustic Fatigue** is associated to high cycle vibrations. The cyclic sound pressure oscillations at frequencies starting at around 50 Hz produce dynamic structural effects associated with inertia and damping.

Typical structural components that are sensitive to acoustic fatigue:

- Skins and panels like those of the fuselage, wing
- and fairings including their supports.

## 10.3 Fatigue Damage

## Fatigue Damage

It would be natural to define a fatigue damage by the density and the number of micro-cracks, and by their interaction.

Before fatigue, the micro-structure has not been yet modified (no dislocation displacements). At this stage, the damage is null.

Once a main crack has propagated resulting in the ruin of the element, the damage is maximum. By convention, it is equal to 1.

This way of defining a fatigue damage is indeed pertinent as it physically represents the micro-structure alteration. But, it is not convenient as it does not allow simple fatigue damage calculations to be done: a model representing all significant micro-structure phenomena would be quite complex and, in fact, not applicable.

## Fatigue Damage

Thus, it is more straightforward to use an experimental way to define a fatigue damage.

Consider a metallic part (test coupon):

- Fatigue cycles are applied up to failure
- Several coupons are tested and broken



## Fatigue Damage

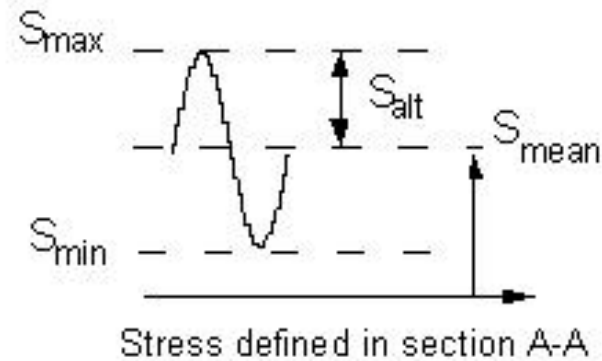
The number of cycles at failure (fatigue life) is not the same for different coupons:

- a large distribution of fatigue lives is obtained for the same fatigue loading
- average value  $N_{\text{mean}}$

$$R = \frac{S_{\min}}{S_{\max}}$$

$$S_{\text{alt}} = \frac{S_{\max} - S_{\min}}{2} = S_{\max} \frac{1 - R}{2}$$

$$S_{\text{mean}} = \frac{S_{\max} + S_{\min}}{2} = S_{\max} \frac{1 + R}{2}$$



## Fatigue Damage

There is no single result per  $S_{max}$  level, but a distribution of fatigue lives  $N$ . It is generally admitted that this distribution follows a logNormal law or  $\log(N)$  follows a Normal law).

The scatter inherent to fatigue must be taken into account for design. It is generally done by applying a Safety Factor to mean fatigue lives.

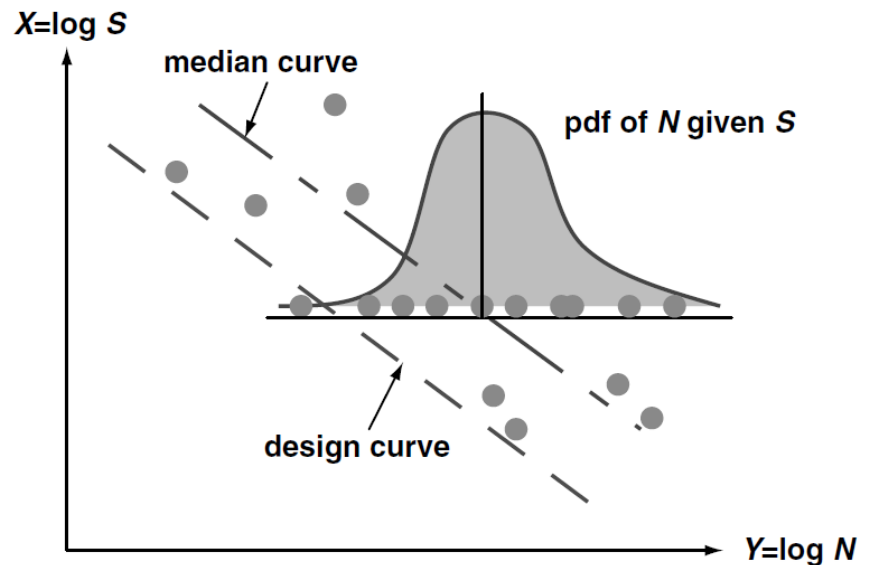
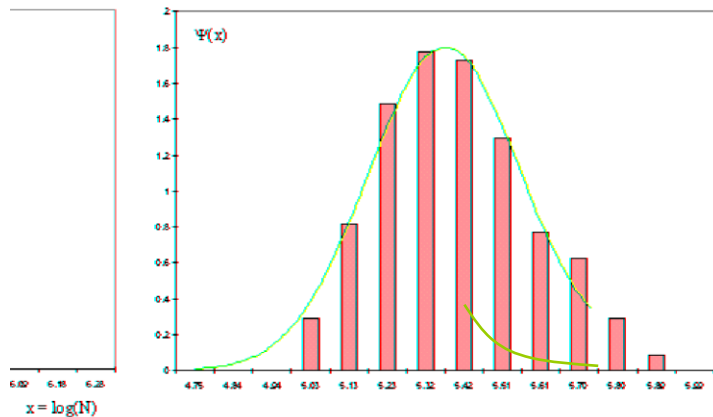


FIGURE 4.5 Concept of a design  $S-N$  curve.

## SN curve

As previously explained, there are several parameters that influence the Fatigue: the geometry, the shape and magnitude of the loads, the material used, and others.

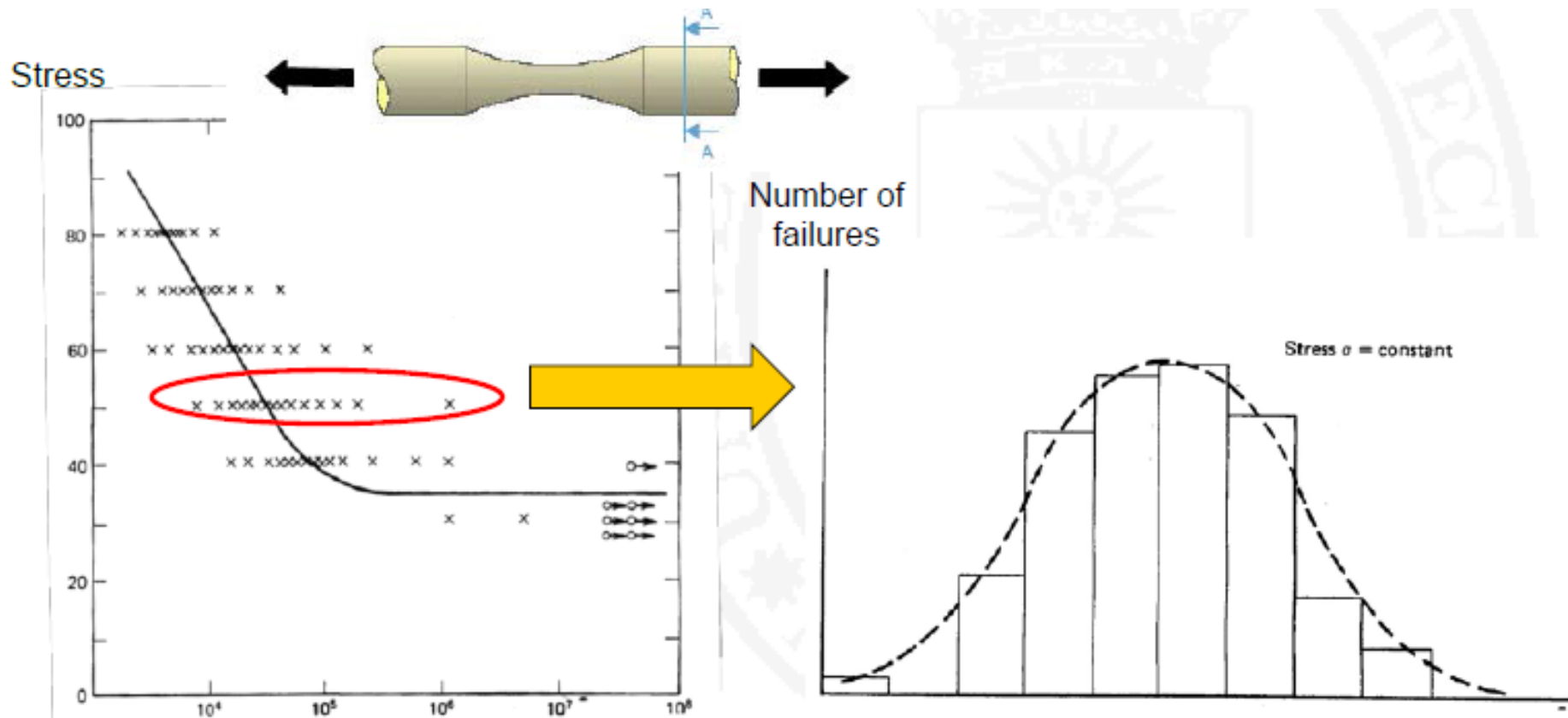
This influence is represented by the so called S-N Curves (or 'Wohler' curves). They indicate the number of cycles (N) with a given shape and stress (S), required to produce a Fatigue crack on a piece of a given material and geometry.

The construction of the S-N Curves is based on tests performed on test pieces that are representative of the material with notches used to represent the concentration of stresses due to the different geometries that can exist. These test pieces are subjected to Fatigue load cycles up to breakage.



## SN curve

The fatigue life of a *structural detail* is represented in the form of a constant amplitude *SN-curve* obtained from tests.



## SN curve - Two examples of SN curves

The fatigue life of a component is represented in the form of a constant amplitude SN-curve ("Wöhler curve") or in form of a "constant life" diagram ("Haigh diagram") in which line of constant endurance  $N$  are plotted on a scale of  $\sigma_a$  against  $\sigma_m$

$\sigma_m$  = mean stress

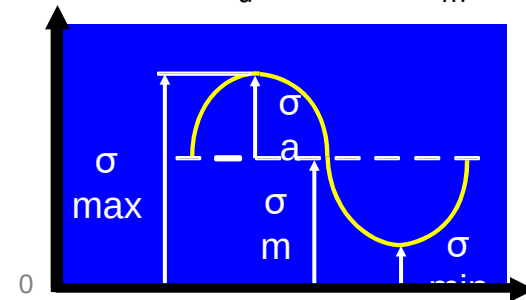
$\sigma_a$  = alternating stress

$\sigma_{max}$  = maximum stress

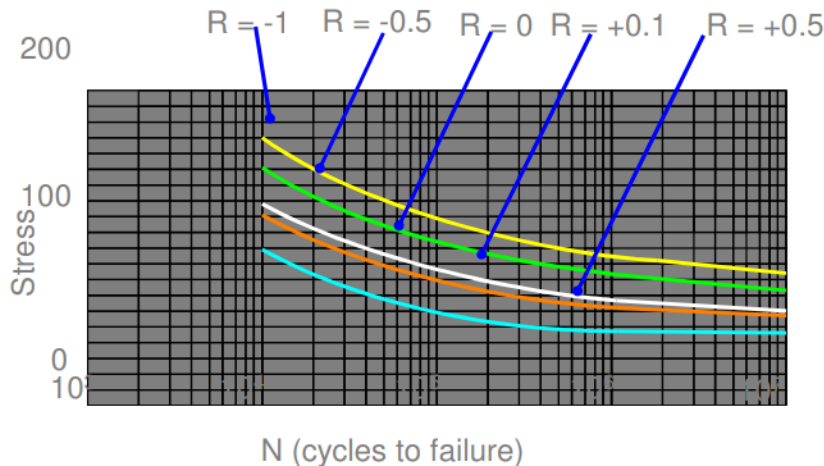
$\sigma_{min}$  = minimum stress

$R$  = stress ratio =  $\sigma_{min}/\sigma_{max}$

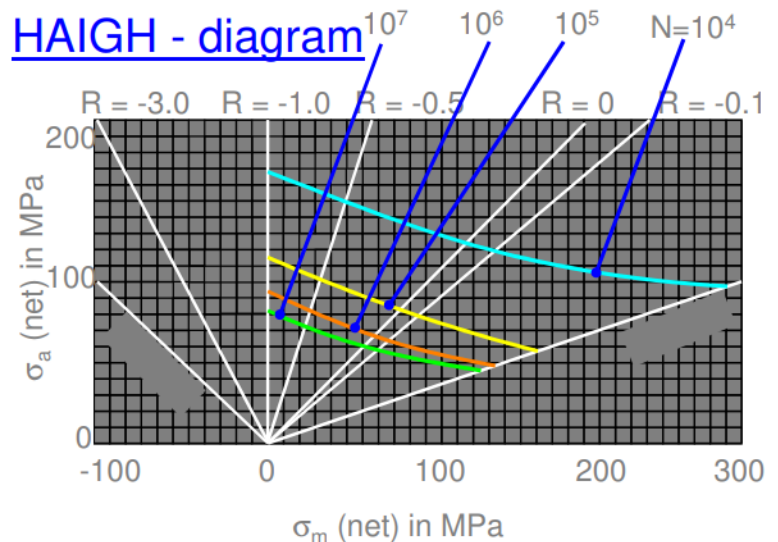
Example for  
 $K_t = 3.0$



### S-N - diagram



### HAIGH - diagram

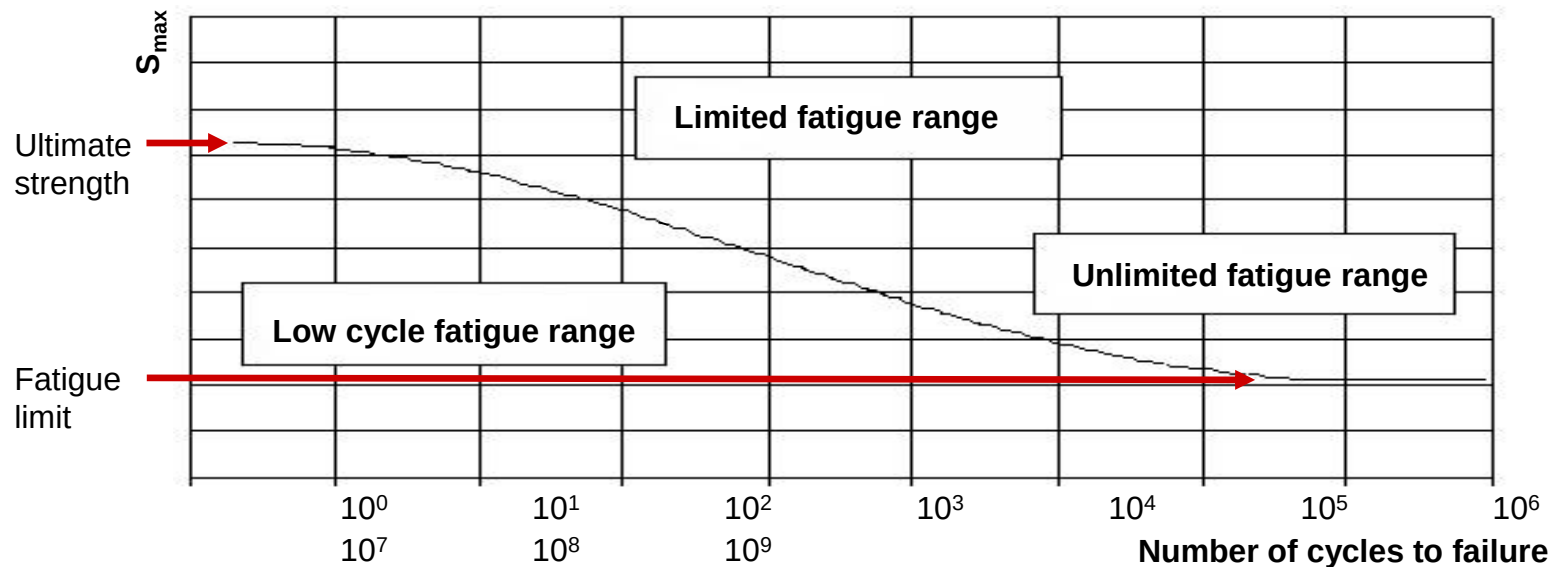


## SN curve

The first domain is characterized by a large material plasticity (low number of cycles)

The second one is approximately between  $10^4$  and  $10^6$  cycles. It corresponds to aeronautical applications.

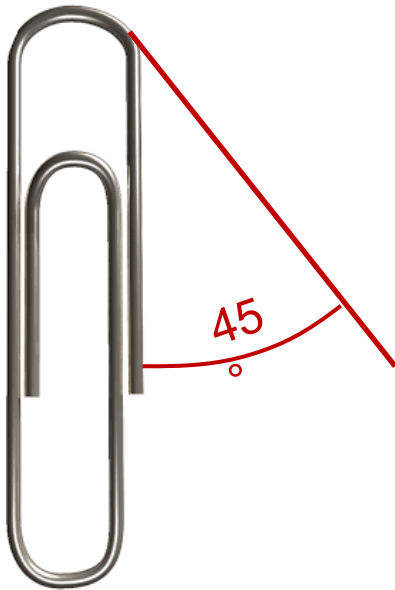
The third one is the unlimited fatigue range. Below a certain stress level, the structure will not fail irrespective of the number of cycles. There is no fatigue damage, but it requires heavy and thick structures to keep the stress low.



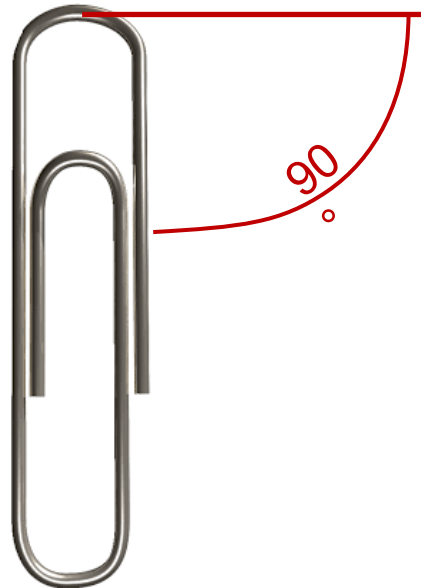
Let's do a test



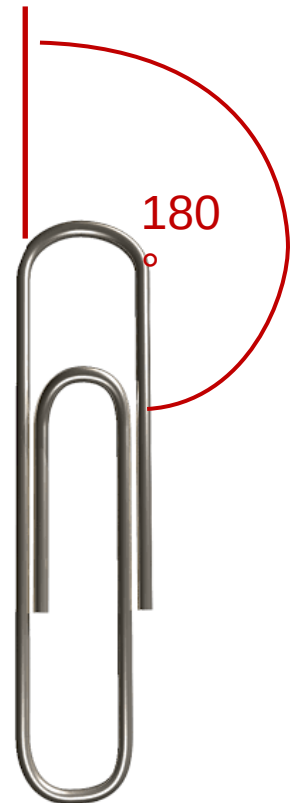
Group 1:



Group 2:



Group 3:



## Damage calculation Miner law

The most used law for the calculation of Fatigue damage of a structural component is the so called “Palmgren-Miner”

It is supposed that the damage suffered by a loaded element at a stresses level of  $\sigma_i$  accumulates linearly with the number of cycles of that stresses level. So,  $N_i$  is the fatigue life corresponding to  $\sigma_i$ , the damage caused by  $n_i$  cycles will be:

$$d_i = n_i / N_i$$

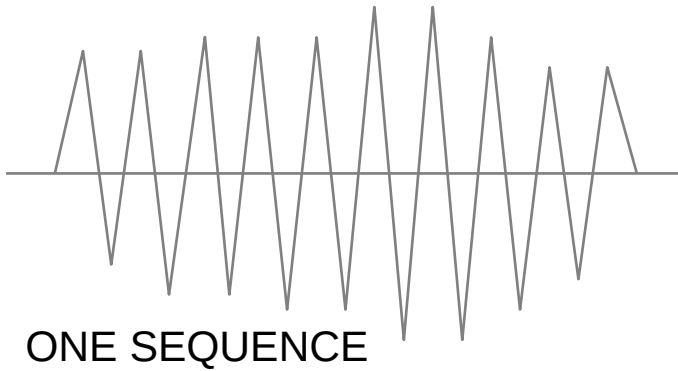
The accumulated damage law of Palmgren-Miner establishes that the total damage for a spectrum of loads formed by cycles with different amplitudes is the sum of the damages of each cycle:

$$D = \sum n_i / N_i$$

So, it is supposed that the failure of the element (the initiation of the first crack) occurs when the value of the total damage exceeds a critical value that usually is 1:

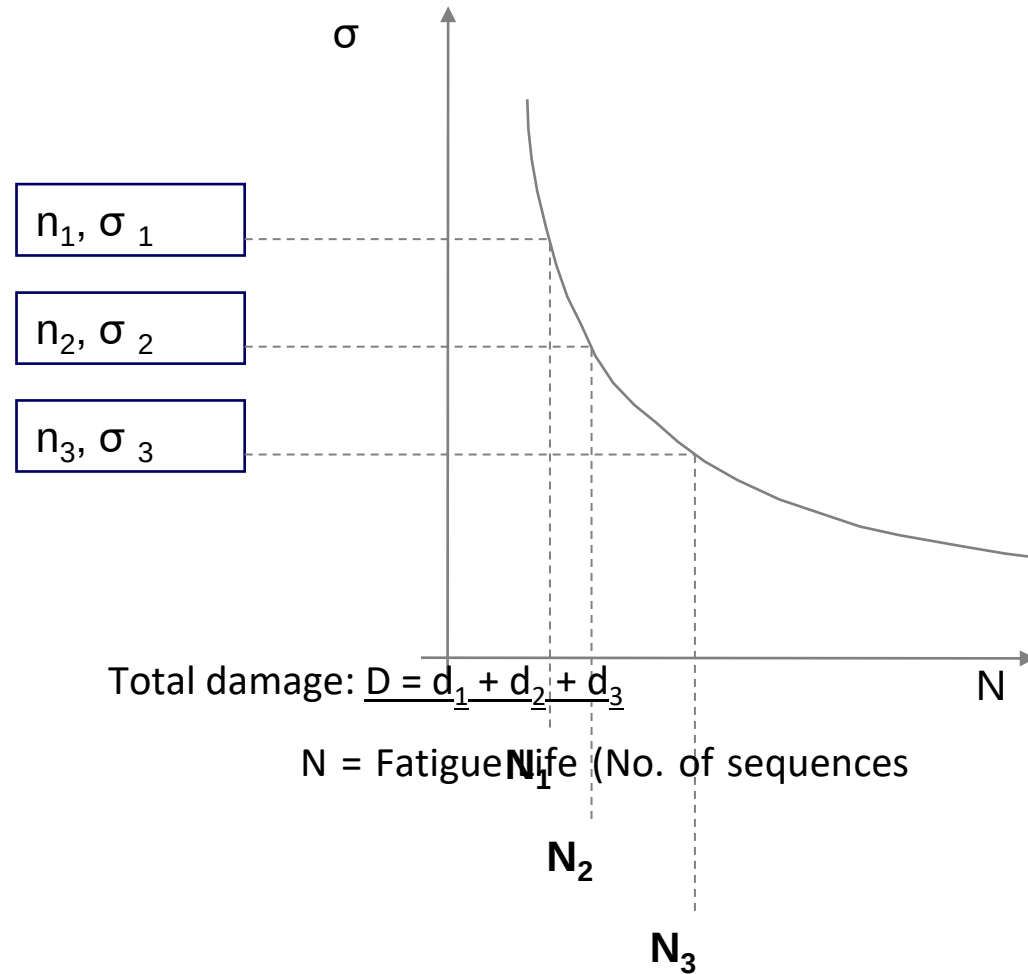
$$\sum n_i / N_i = 1 \Rightarrow \text{Fatigue crack}$$

## Damage calculation Miner law



$$\left. \begin{aligned} d_1 &= n_1 / N_1 \\ d_2 &= n_2 / N_2 \\ d_3 &= n_3 / N_3 \end{aligned} \right\} \rightarrow \text{up to failure):}$$

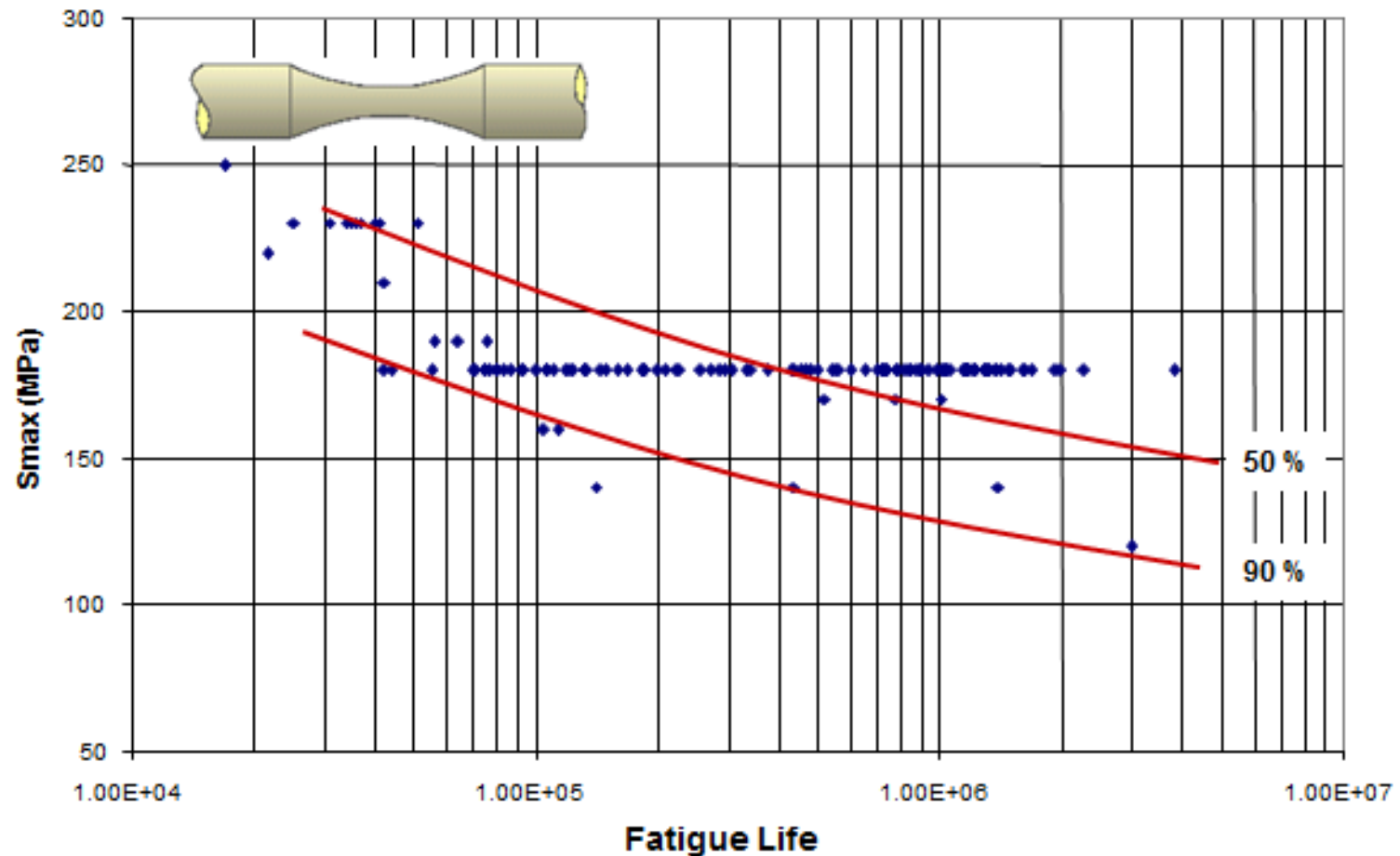
$$N = 1 / D$$



## 10.3 Exercises

## Exercise 1

On the graph below, several Smax levels were tested with a R ratio equal to 0:



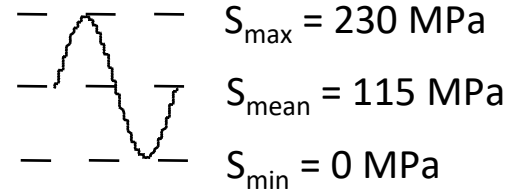


**Exercise 1**

Consider 2 different successive fatigue loadings:

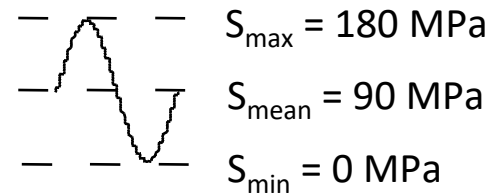
– Loading n°1

$n_1$  cycles resulting in damage  $D_1 = n_1 / N_{\text{mean1}}$   
 $n_1 = 30\,000$  cycles



– Loading n°2

$n_2$  cycles resulting in damage  $D_2 = n_2 / N_{\text{mean2}}$   
 $n_2 = 50\,000$  cycles

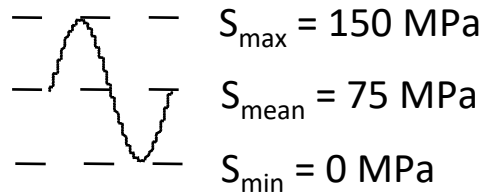


What the total damage?

**Exercise 2**

A third fatigue loading is applied ( $S_{\max}=150$  MPa,  $R=0$ ).  
How many cycles  $n_3$  are expected at failure ?

$$\text{At failure } D_{\text{total}} = 1 \Rightarrow n_1/N_{\text{mean1}} + n_2/N_{\text{mean2}} + n_3/N_{\text{mean3}} = 1$$
$$n_3 = [1 - (n_1/N_{\text{mean1}} + n_2/N_{\text{mean2}})] \cdot N_{\text{mean3}}$$

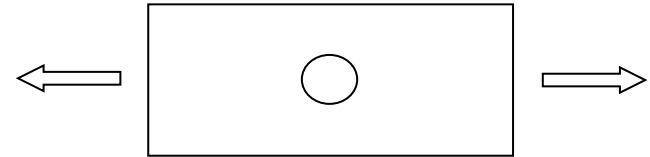


$N_{\text{mean1}}$ ,  $N_{\text{mean2}}$ ,  $N_{\text{mean3}}$  can be read on so-called « S-N curves » (or « Wöhler curves »)

### Exercise 3

Let's consider a plate with an open hole.

- Width: 50 mm
- Hole diameter: 10 mm
- Thickness: 3 mm



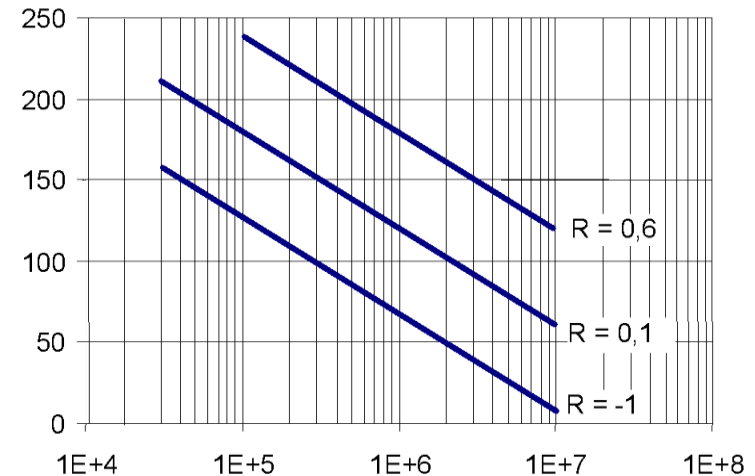
We have performed two series of loading :

- $F1(t) = 12000 + 9840 \cdot \sin(t)$  (Unit : Newton)
  - $n1 = 30\,000$  cycles
- $F2(t) = 9600 + 7860 \cdot \sin(t)$ 
  - $n2 = 70\,000$  cycles

We want to continue with another loading :

- $F3(t) = 7200 + 5880 \cdot \sin(t)$

How many cycles  $n$  could we perform ?



1. Hans-Jürgen Schmidt. “Airbus Fatigue & Damage Tolerance Course”, 2005
2. Airbus training (ESAD). “Fatigue and Damage Tolerance Introduction Course”, 2005
3. Airbus Military (TEASA). “Durability and Damage Tolerance (DaDT)- P23 Indirect Offset”, June 2017.
4. R.E. Peterson. “Stress Concentration Factors”. 2<sup>nd</sup> Edition) – 1997
5. MIL-HDBK-5H, 1 December 1998